

Fluorescence Spectroscopy

Firfly
↓
Orange → (560-600nm)
Luciferin

(photo)Luminaeance

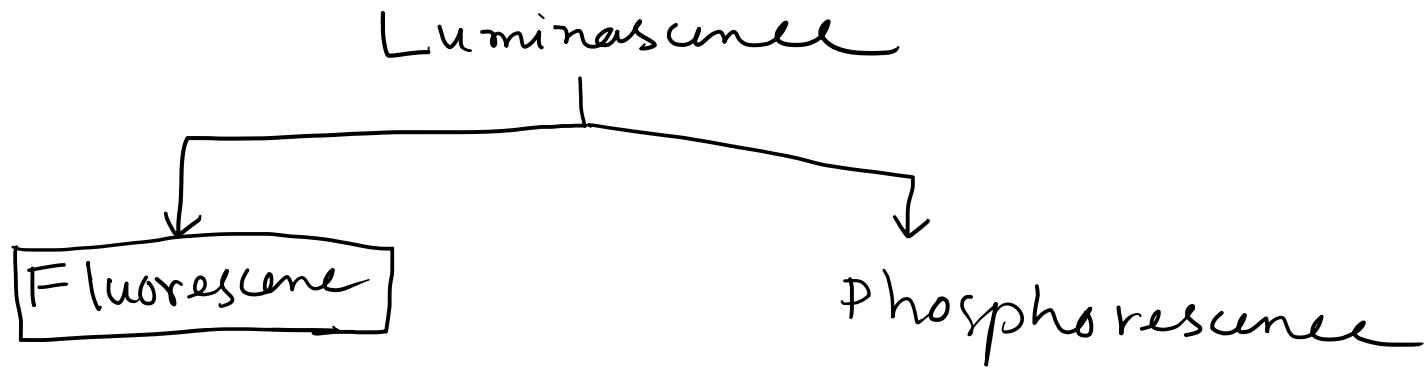
Def.: Spontaneous emission of light/
Signal due to chem. reaction or pressure/structural change

Chemilaseance
(Because of chemical Reaction)

- $\text{luminol} + \text{H}_2\text{O} \xrightarrow{\text{Fe}} \text{Light}$
- many

Bio-luminaeance

- Jellyfish (GFP)
- Firefly



Fluorescence: Instant re-emission of light from Siglet-state.

- The ^{time taken}_A in Absorption of light = 10^{-15} s

$$= \boxed{\text{fs}}$$

- Emission of light happens due to fluorescence in

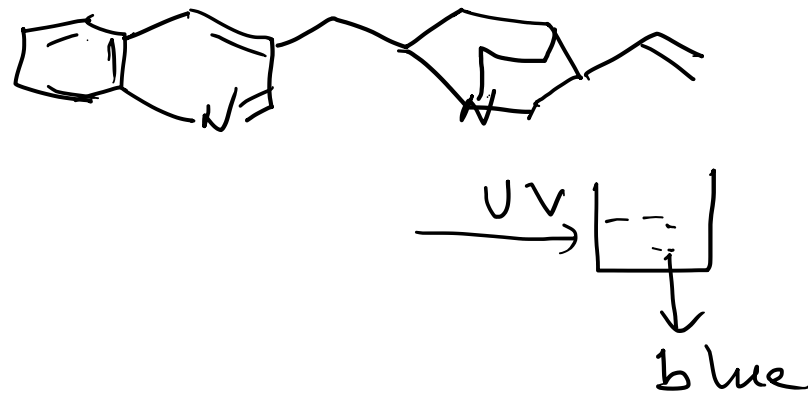
$$10^{-9} \text{ s}$$

$$= \boxed{\text{ns}}$$

Phosphorescence : Delayed emission of light/signal from Triplet state.

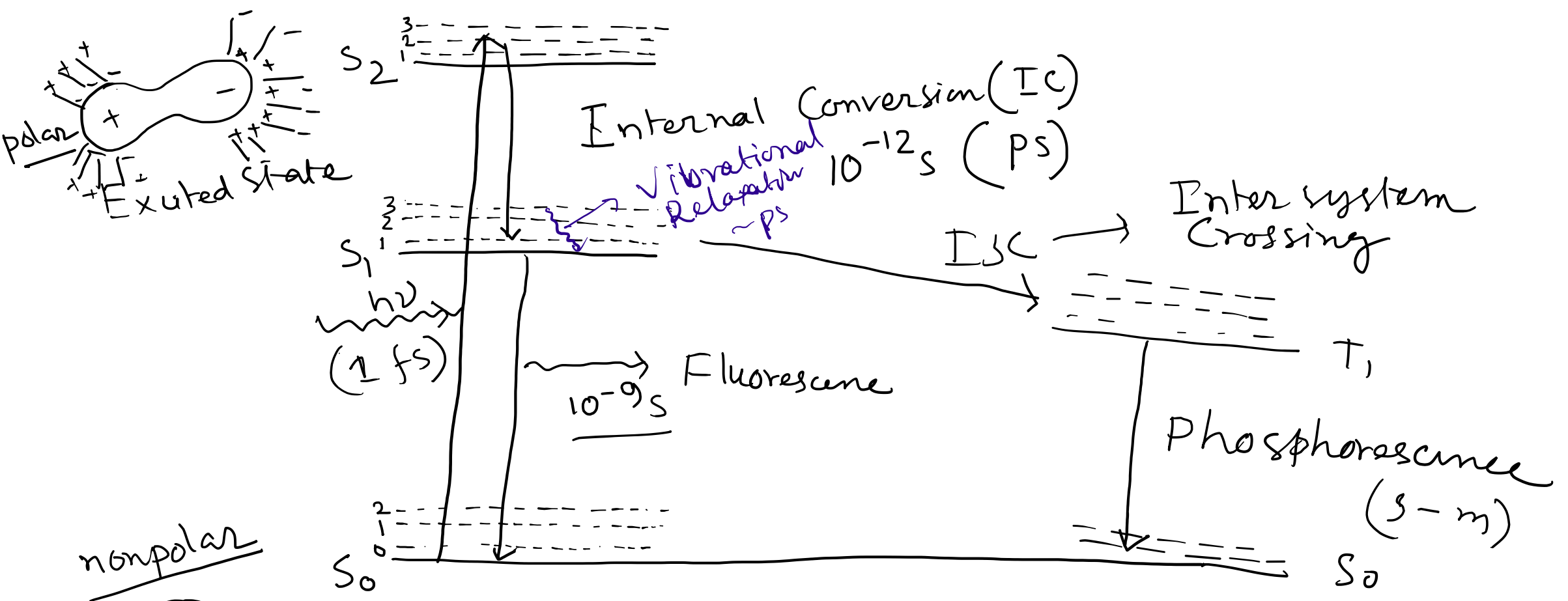
Fluorophore : Aromatic molecule

↓
Quinine (medicine of malaria)



Rearrangement of solvent
is known as Solvation dynamics

Jablonski Diagram



Fluorescence Spectroscopy deals with
excited state reaction/rearrangement/energy transfer.

Characteristics of Fl. Em.

① Stokes shift

$$\Delta E_{\text{Abs}} > \Delta E_{\text{em}}$$

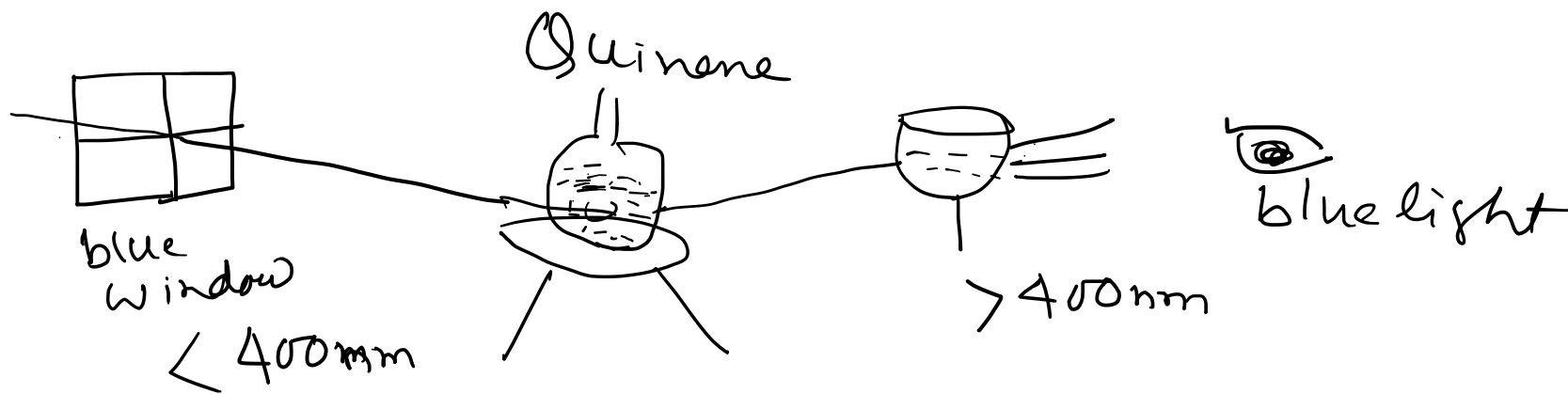
$$E = h\nu \\ = h \frac{c}{\lambda}$$

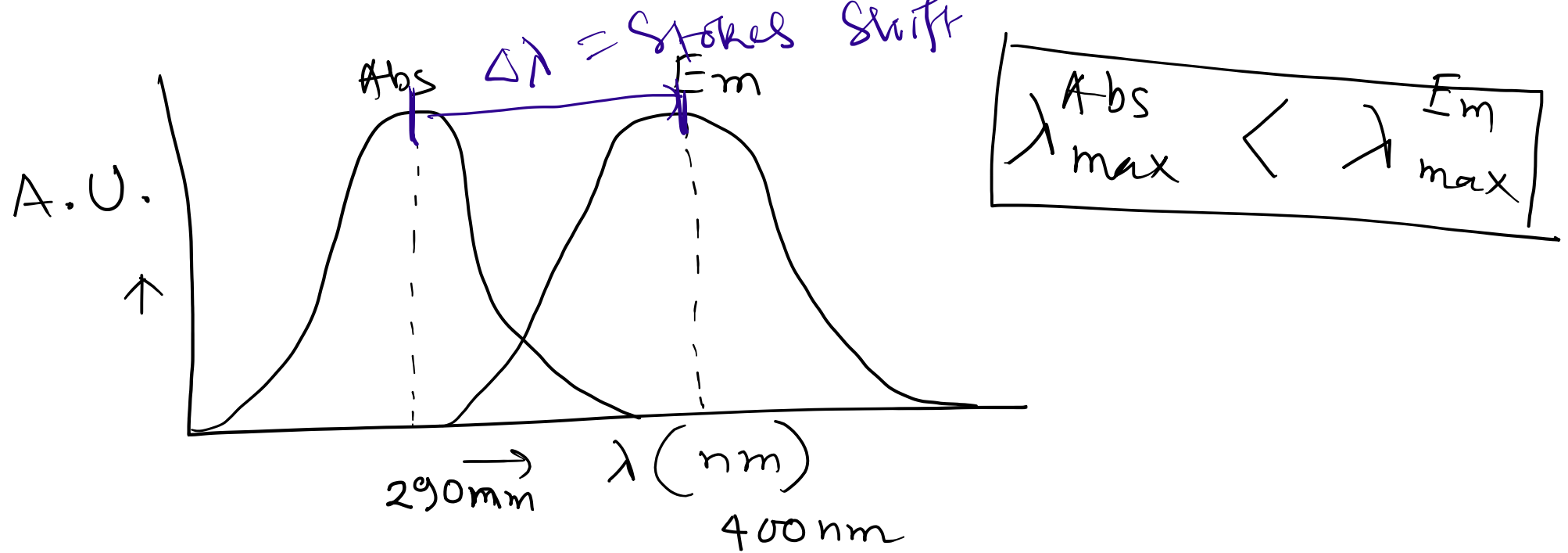
$$\lambda_{\text{Abs}} < \lambda_{\text{em}}$$

$$\Delta E_{\text{em}} = E_{S, \nu_0} - E_{S, \nu_1}$$

$$\Delta E_{\text{Abs}} = E_{S_2/S/\nu_h} - E_{S, \nu_0}$$

$E_{\text{em Max}}$ is always higher than Abs. maxima

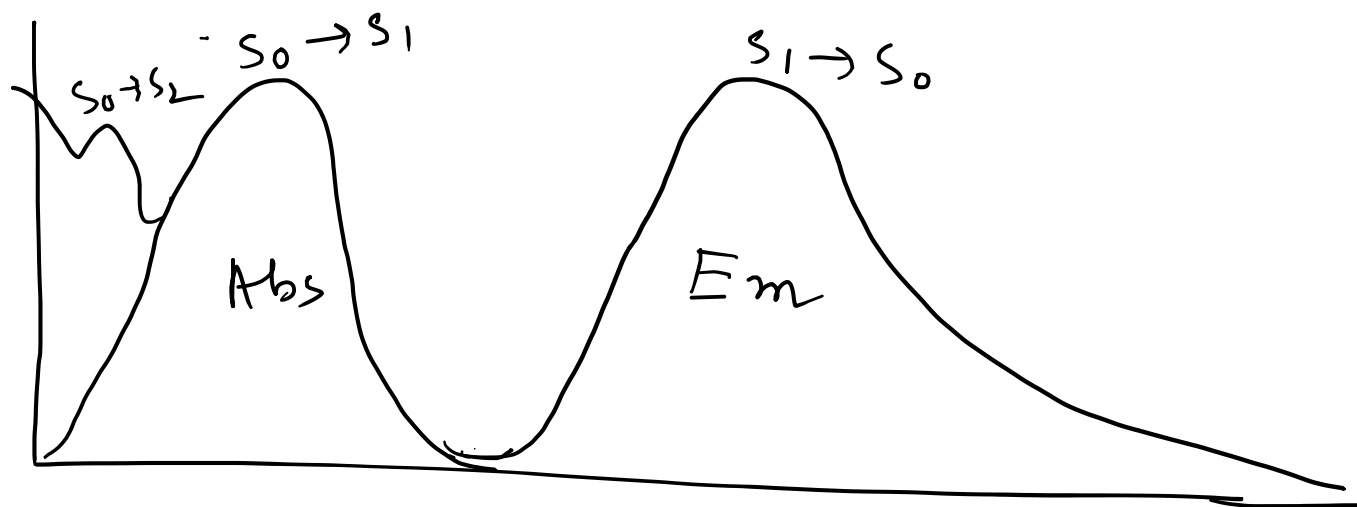
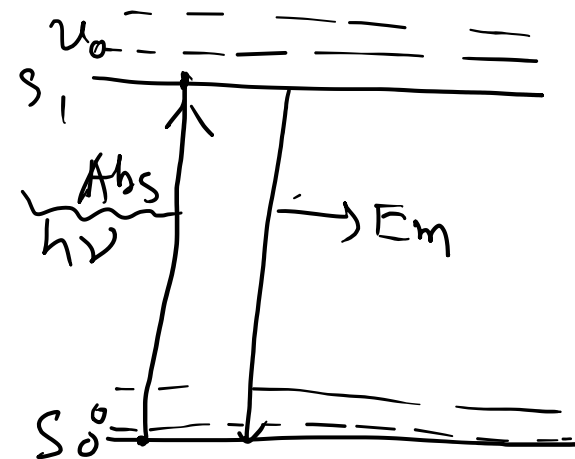
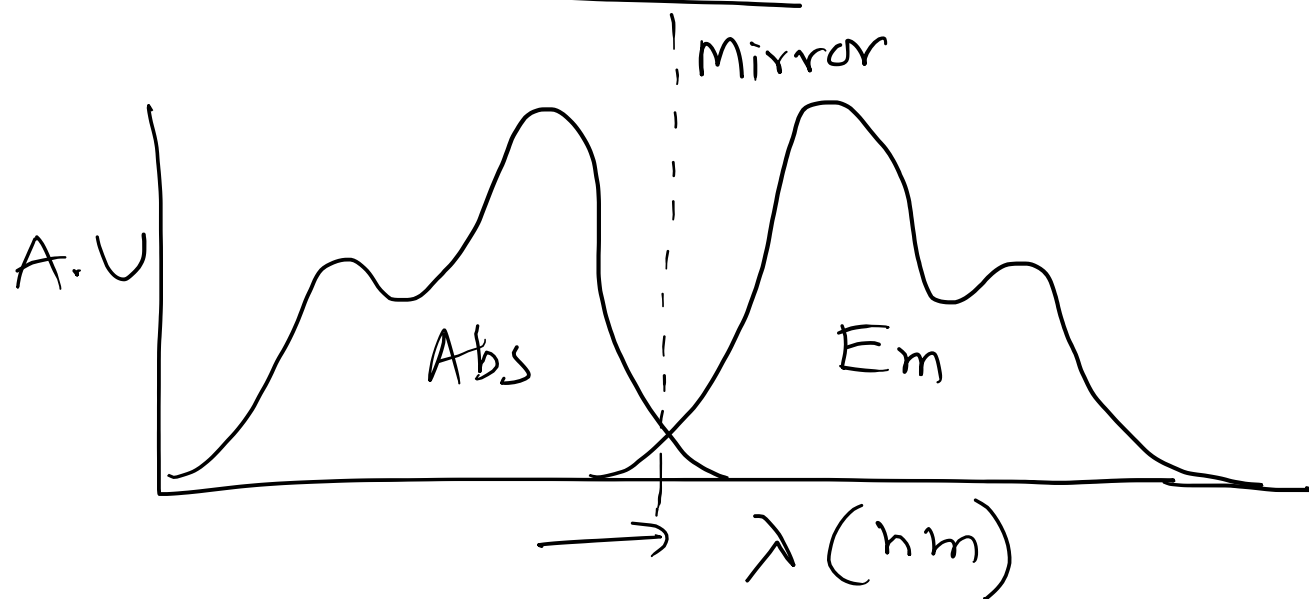




② Kasha's Rule: Emission spectra are typically independent of excitation spectra

③

Mirror Image Rule



Applications

- ① Biotech Industry
- ② Flow Cytometry
- ③ Medical dia.
- ④ DNA/ RNA Ana
- ⑤ Genetic
- ⑥ Basic Bioscience Res.

Fluorescence lifetime: Average time the molecule spends in the excited state before returning to the ground state

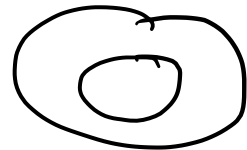
depending on solution/structure

(τ)

1 ns in water

100 ps

t = real-time



micelle/vesicle

~ 1000 ns

1 - 10 fs

Slow Component

Fast Component

Quantum Yield

$$Q.Y. = \frac{\text{No. of photon emitted}}{\text{No. of photon Absorbed}}$$

